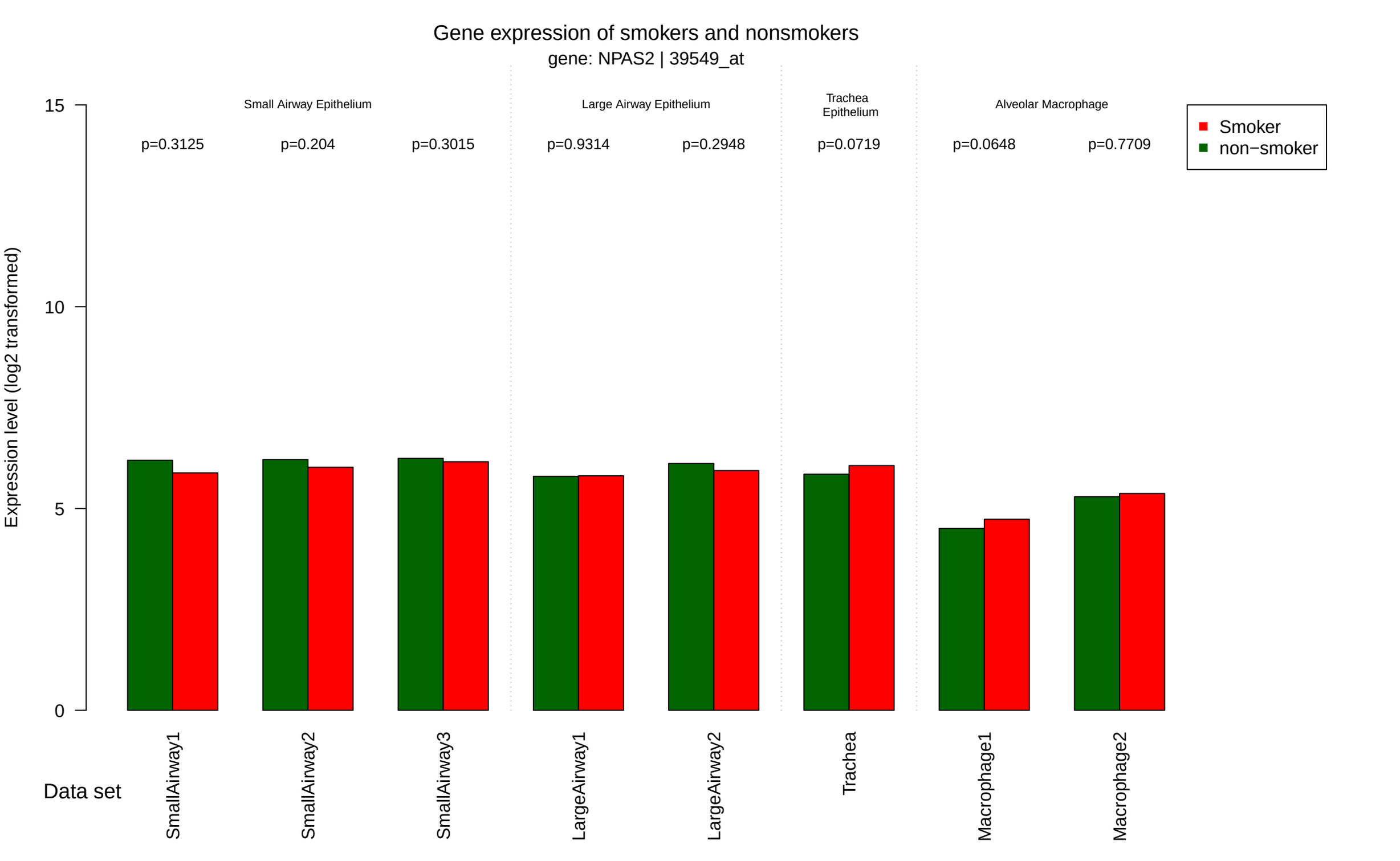
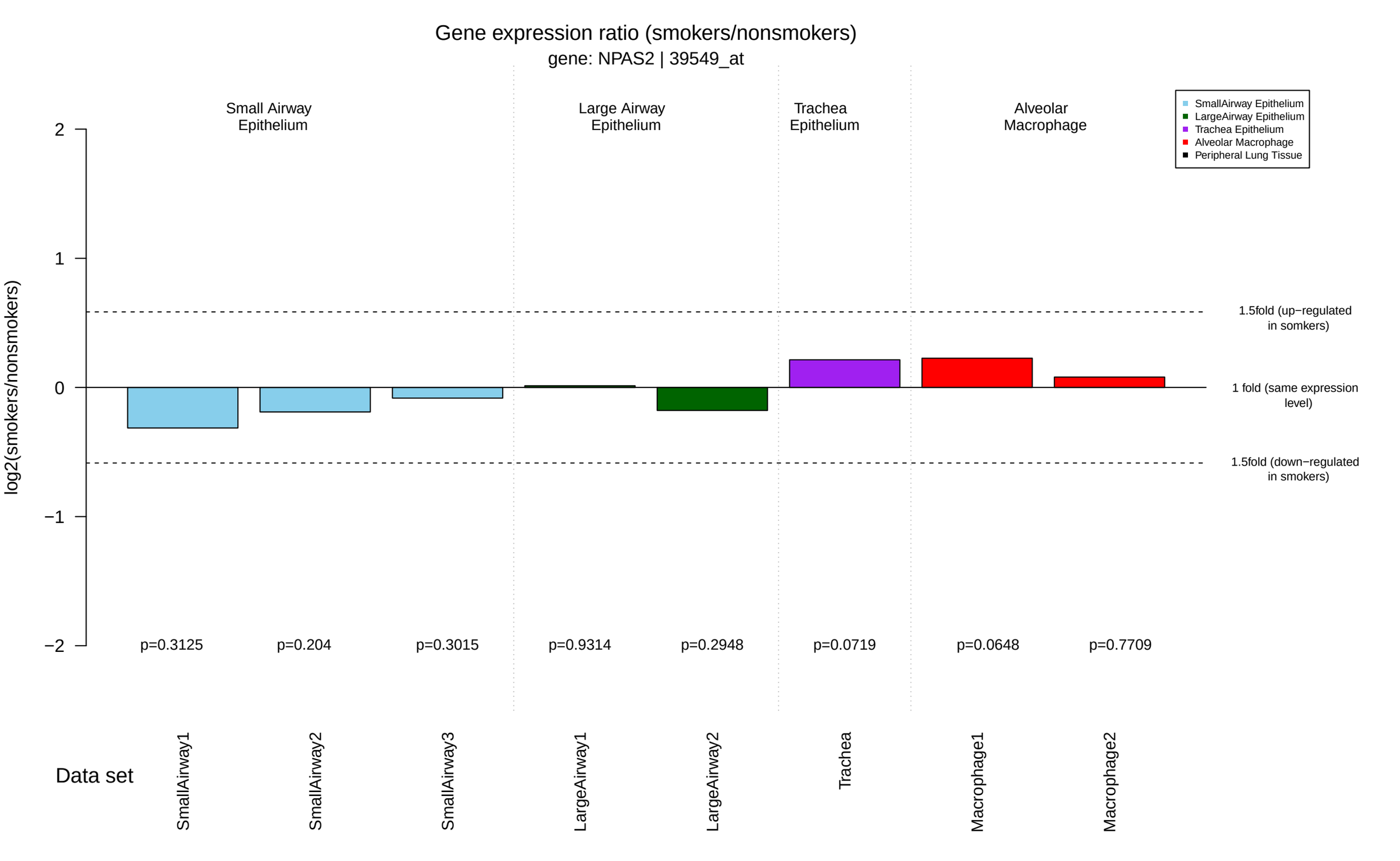
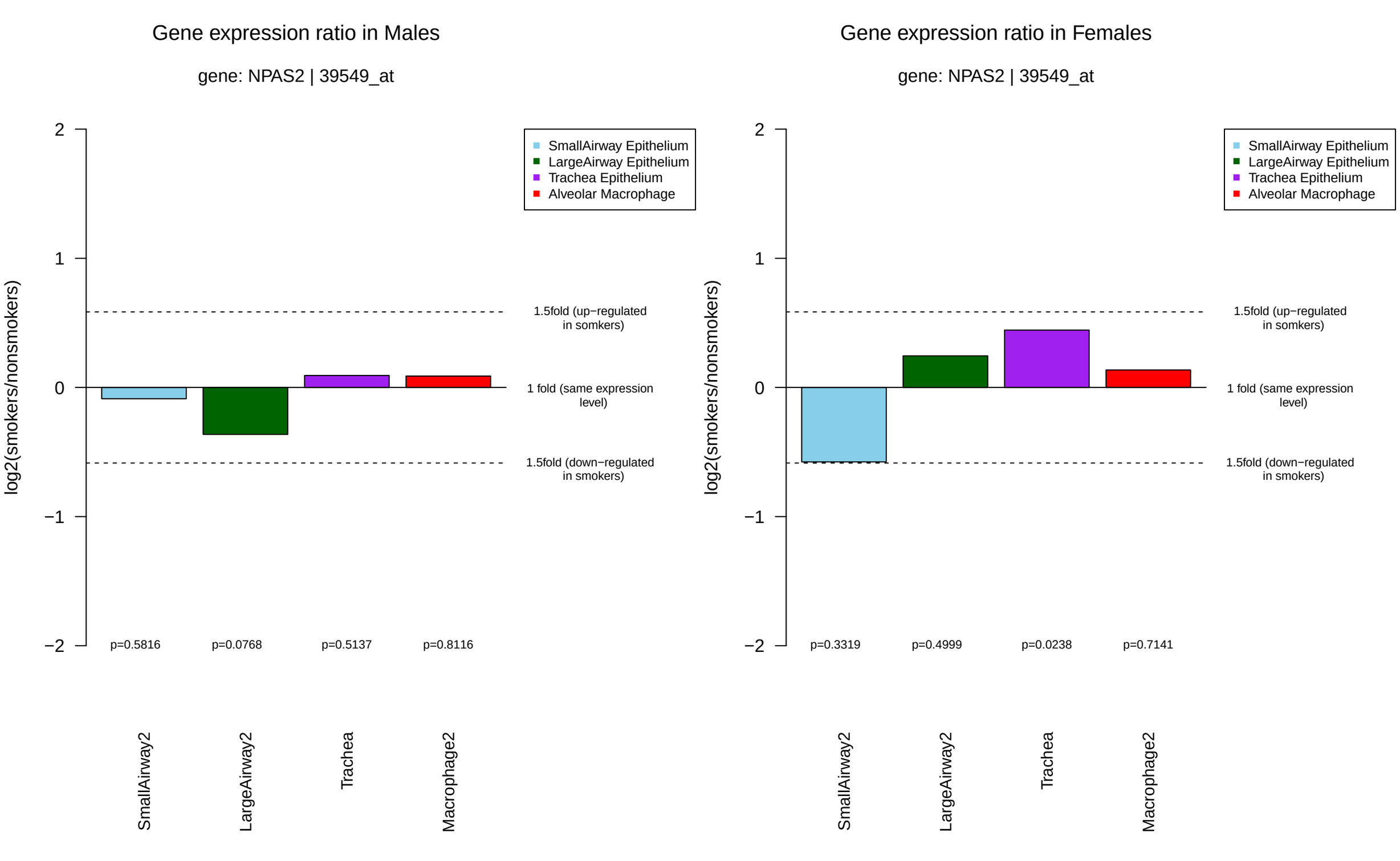
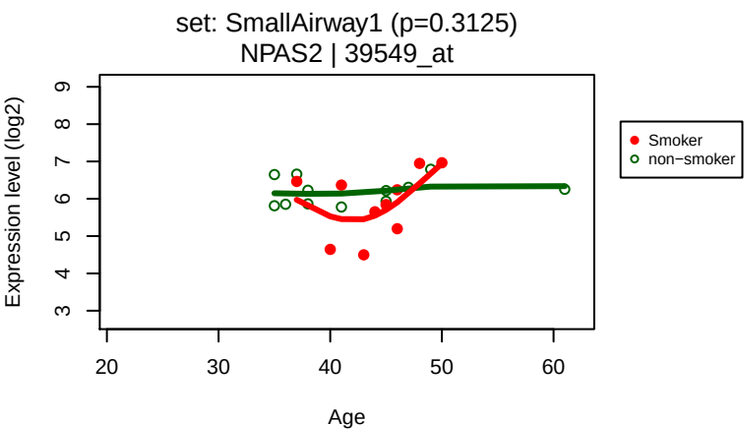
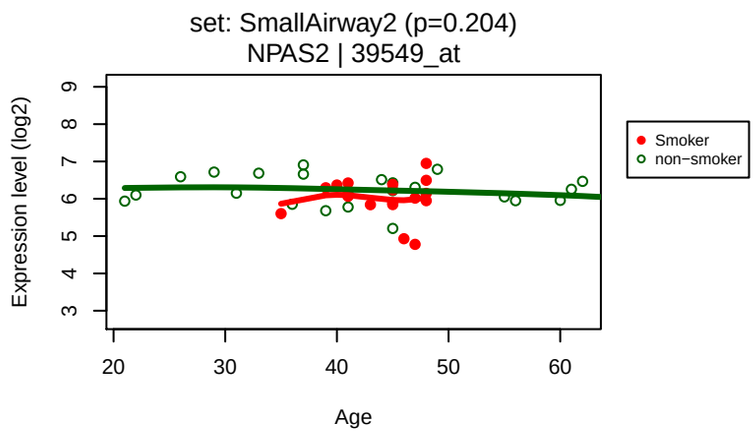
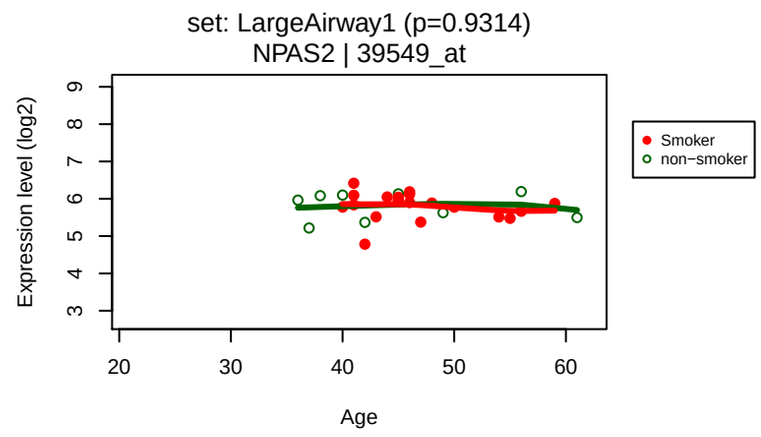

Gene expression of smokers and nonsmokers
gene: NPAS2 | 39549_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3125
p=0.204
p=0.3015
p=0.9314
p=0.2948
p=0.0719
p=0.0648
p=0.7709
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

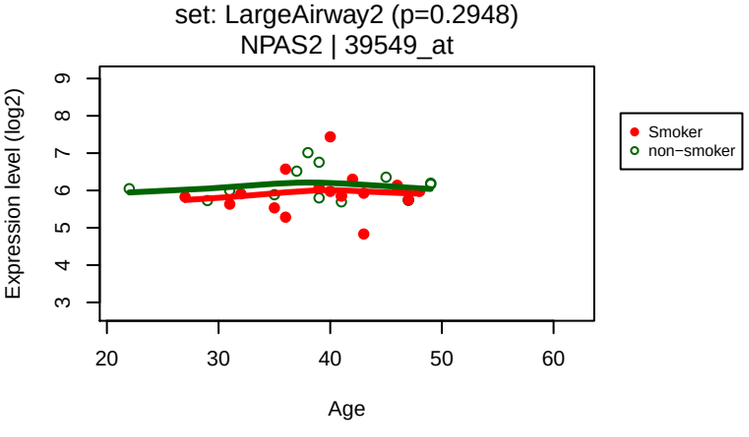

Gene expression ratio (smokers/nonsmokers)
gene: NPAS2 | 39549_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3125
p=0.204
p=0.3015
p=0.9314
p=0.2948
p=0.0719
p=0.0648
p=0.7709
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

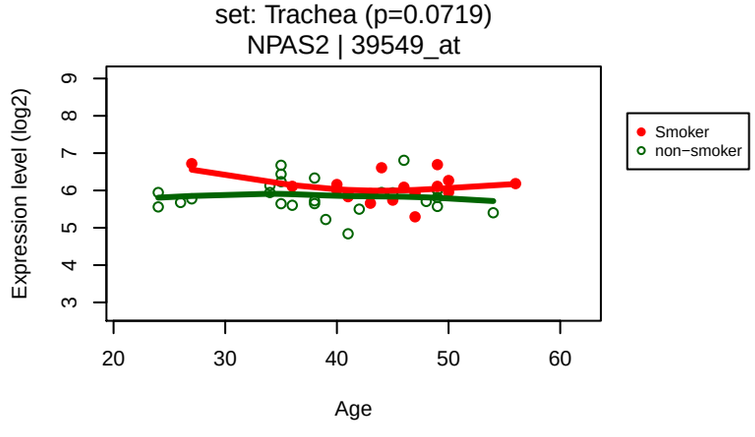

Gene expression ratio in Males
gene: NPAS2 | 39549_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5816
p=0.0768
p=0.5137
p=0.8116
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPAS2 | 39549_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3319
p=0.4999
p=0.0238
p=0.7141
SmallAirway2
LargeAirway2
Trachea
Macrophage2

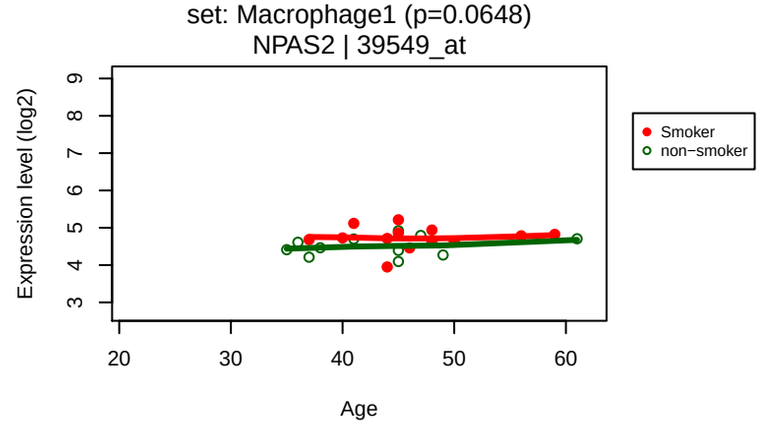

set: SmallAirway1 (p=0.3125)
NPAS2 | 39549_at
Expression level (log2)
Age
Smoker
non−smoker

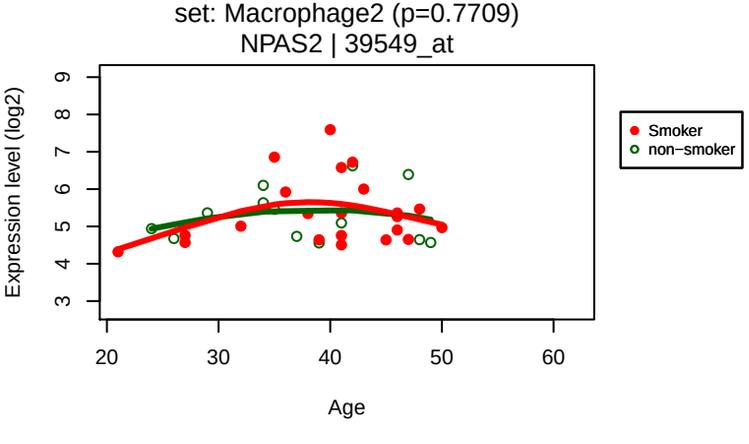

set: SmallAirway2 (p=0.204)
NPAS2 | 39549_at
Expression level (log2)
Age
Smoker
non−smoker

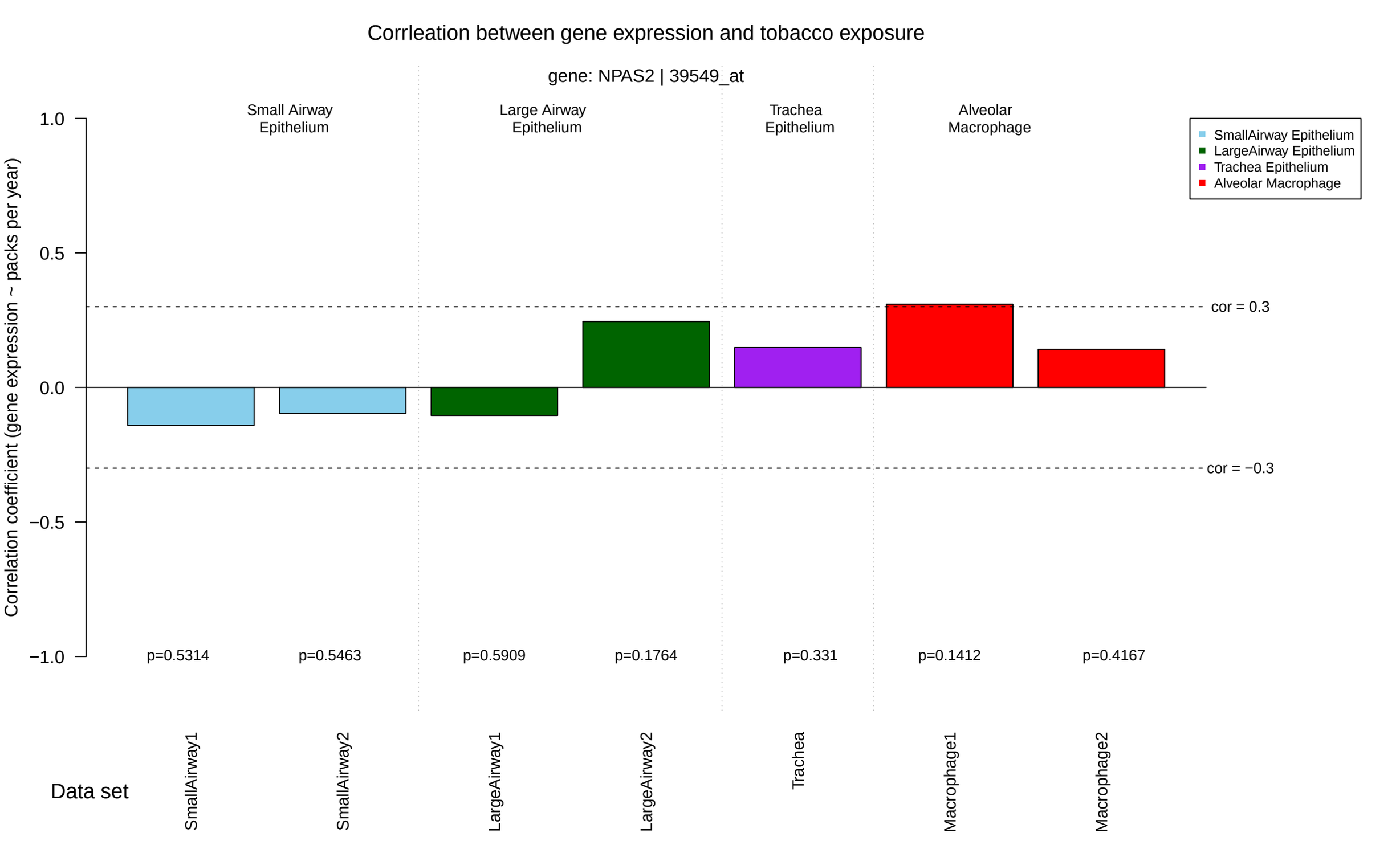

set: LargeAirway1 (p=0.9314)
NPAS2 | 39549_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2948)
NPAS2 | 39549_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0719)
NPAS2 | 39549_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0648)
NPAS2 | 39549_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7709)
NPAS2 | 39549_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPAS2 | 39549_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5314
p=0.5463
p=0.5909
p=0.1764
p=0.331
p=0.1412
p=0.4167
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2